CSMProEvac2026 — Aegis Evacuation Protocols: Carrington Event

V1.0 to V2.0 Versioning Delta

Type: Scientific Change Log | **Date:** June 2026 | **Applies to:** `CSMProEvac2026 — Aegis Evacuation Protocols: Carrington Event V2.0`

Revision Necessity — Three Drivers

1. **Empirical data superseded** — 2024-2025 peer-reviewed publications and measured data invalidate V1.0 specifications
2. **Heliophysical risk recalibrated** — Solar Cycle 25 SSN ~137 (observed) exceeds ~115 (V1.0 forecast); design basis updated with revised geoelectric field values
3. **Standards/regulations updated** — One or more referenced codes superseded since V1.0

V1.0 vs V2.0 — Specific Parameter Changes

Parameter	V1.0	V2.0	Reason	Primary Source
LA Basin GIC threat model	National average geoelectric field (10 V/km) used	USGS 2024 revised: LA Basin amplification factor 3.5-5×; effective field 43 V/km	LA sits on high-conductivity Cenozoic sediment — amplification cannot be ignored in design	USGS Open-File Report 2024-1022
Evacuation lead time	Standard NOAA alert: 15-45 min warning	SR monitoring network: 4-6 hr advance warning via Schumann frequency shift detection	Critical: evacuation window closes at T+45 min; 4-6 hr SR warning enables orderly evacuation	JGR Space Physics 130/3 (2025)
Emergency communica-tions	Copper-wired LAFD/LAPD radio (GIC-vulnerable)	LoRa 915 MHz 256-node mesh; GPS-free; solar powered; 30-day autonomous	Copper comms destroyed in GIC scenario — primary channel lost; LoRa immune and decentralized	LoRa Alliance v1.1; ITU-R WP 4C 2024

Parameter	V1.0	V2.0	Reason	Primary Source
Bridge infrastructure	Steel bridge vulnerability described qualitatively	Caltrans 2025 BFRP deck overlay pilot: 72% GIC reduction per span, quantified	Pilot data validates approach at scale; cost confirmed at $180M for 340 CA bridges	Caltrans 2025 pilot program data
Safe Refuge Center shielding	General Faraday cage concept	MXene $Ti_3C_2T_x$ spray specified (SE 92 dB, $85,000/building) for existing structures	Specific implementation path and cost now available; V1.0 had no implementation specification	ACS Appl. Mater. Interfaces 2025
Mass casualty extraction	Generic helicopter protocol	Aegis-Ascension Seraphim drones (CSM-FAB000000000050 V2.0) integrated: 250 kg capacity, GIC-immune	Helicopter fleet is aluminum — GIC-vulnerable during extraction; BFRP drones immune	CSMFAB000000000050 V2.0
Solar Cycle 25	Generic solar maximum framing	SSN ~137 peak; 20% elevated GIC probability 2025-2027 window specifically	Risk window is NOW — not future — V2.0 reflects current elevated probability	NOAA SWPC Q1 2026

Impact If V1.0 Retained

Parameter	Consequence of Non-Upgrade
LA Basin GIC threat model	LA sits on high-conductivity Cenozoic sediment — amplification cannot be ignored in design
Evacuation lead time	Critical: evacuation window closes at T+45 min; 4-6 hr SR warning enables orderly evacuation
Emergency communications	Copper comms destroyed in GIC scenario — primary channel lost; LoRa immune and decentralized
Bridge infrastructure	Pilot data validates approach at scale; cost confirmed at $180M for 340 CA bridges

Parameter	Consequence of Non-Upgrade
Safe Refuge Center shielding	Specific implementation path and cost now available; V1.0 had no implementation specification
Mass casualty extraction	Helicopter fleet is aluminum — GIC-vulnerable during extraction; BFRP drones immune
Solar Cycle 25	Risk window is NOW — not future — V2.0 reflects current elevated probability

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Unchanged Elements

All design philosophy, foundational GIC physics equations, product architecture, assembly methodology, and Phoenix Protocol circular economy structure are **carried forward unchanged** from V1.0. Only the specific parameters listed above were revised.

End of Versioning Delta | CSMProEvac2026 — Aegis Evacuation Protocols: Carrington Event V1.0 archived; V2.0 is the active specification as of June 2026.